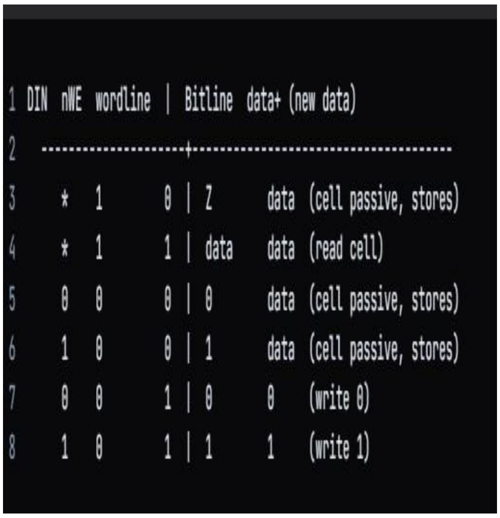
CHAPTER-4
EXISTING SYSTEM

4.1 6T SRAM Cell:

A 6T SRAM (Six-Transistor Static Random Access Memory) cell is a basic building block in semiconductor memory technology. It consists of six transistors, which form two cross-coupled inverters for storing data bits and two additional transistors for accessing the stored data. The 6T SRAM cell is widely used due to its advantages such as high speed, low power consumption, and stability.

The 6T SRAM cell operates by storing data in the cross-coupled inverters, which maintain the stored data as long as power is supplied. The access transistors control the read and write operations by connecting the cell to the bit lines. The input drivers, when activated, can cause a short-circuit condition, leading to the overwriting of the storeddata. However, this is mitigated by the weak transistors inside the SRAM cell.

The following table shows the input and output conditions for the 6T SRAM cell:



DIN	nWE	wordline	Bitline	data+ (new data)
*	1	0	Z	data (cell passive, stores)
*	1	1	data	data (read cell)
0	0	0	0	data (cell passive, stores)
1	0	0	1	data (cell passive, stores)
0	0	1	0	0 (write 0)
1	0	1	1	1 (write 1)

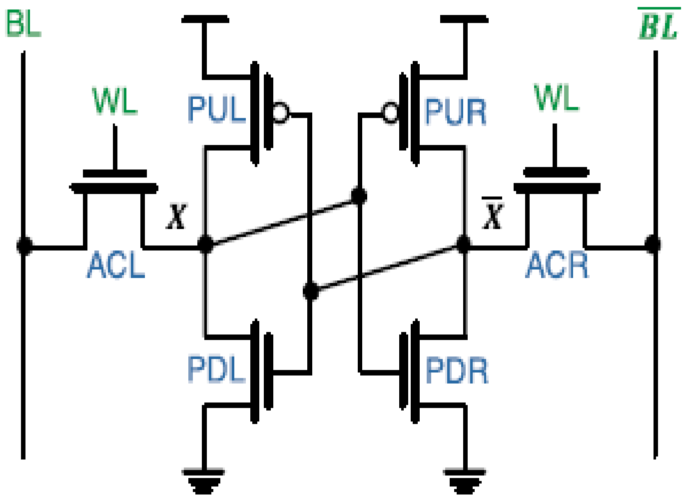


Fig 4.1 schematic of 6T

In summary, the 6T SRAM cell is a fundamental component in semiconductor memory technology, providing high-speed, low-power, and stable data storage. The cell consists of six transistors, which store data using cross-coupled inverters and control read and write operations through access transistors. The input drivers, when activated, can cause a short-circuit condition, but this is mitigated by the weak transistors inside the SRAMcell.

4.2 8T SRAM Cell:

An 8T SRAM cell is a type of static random-access memory (SRAM) cell that uses eight transistors to store a single bit of data. It has improved stability and reduced leakage power compared to the conventional 6T SRAM cell. The 8T SRAM cell consists of two cross-coupled inverters for storing data and two additional access transistors for read and write operations. It is more complex than the 6T SRAM cell but offers better performance and stability.

The main points about the 8T SRAM cell are:

- **Improved stability**: The 8T SRAM cell has better read and write stability compared to the 6T SRAM cell due to the additional access transistors.
- **Reduced leakage power**: The 8T SRAM cell reduces leakage power by using sleep transistors or power gating techniques.
- **Increased complexity**: The 8T SRAM cell is more complex than the 6T SRAM cell due to the additional transistors, which can increase the area and power consumption.
- **Compatibility:** The 8T SRAM cell can be used in both bulk CMOS and FinFET technologies.

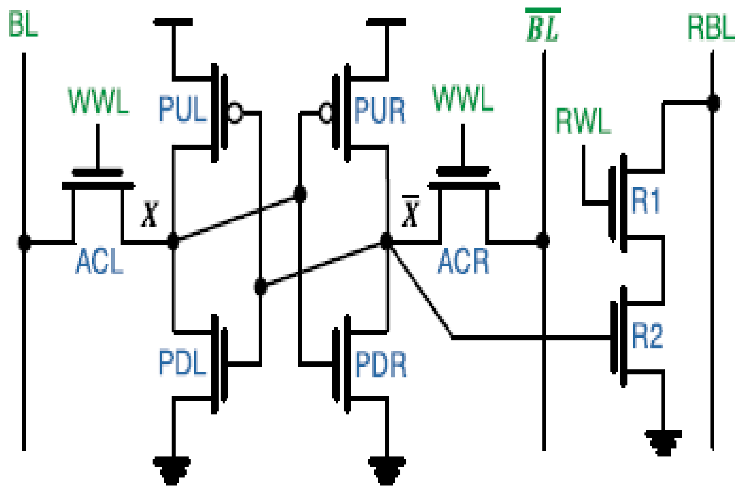


Fig 4.2 Schematic of 8T

4.3 10T SRAM Cell:

SRAM is a cell memory semiconductor. It stores a bit of info. This requires much less fuel than most memory cells and is much quicker. Researchers are interested in further developing SRAM cells because of its robustness and durability. Within a chip or microprocessor IC, SRAM is a critical part. The architecture of a Nanoscale SRAM cell has become a problem because the noise margins are limited and threshold voltage variation is more sensitive. In terms of reliability and durability, 10 T SRAM cell performs better than 6 T SRAM cell. Due to the degradation of noise thresholds, the 6 T SRAM cell has less efficiency at low voltage.

A 10 T SRAM cell with 32 nm CMOS technology is developed using an EDA tanner method. The capacity, delay and power delay characteristics will be analyzed. In fact, 10 T SRAM cell is 6 T SRAM cell with four additional transistors. Additional read circuits are attached to this 10 T SRAM cell to prevent cell flip-out. The 10 T SRAM cell equipped with 180 nm CMOS technology has 22.08 x 10-9 W, 39.95 x 10-9 s and 0.8820 x 10-15 Ws respectively, the power dissipation, delay and power delay element.

The 10T SRAM cell offers several advantages over conventional 6T SRAM cells, including:

- **Improved stability**: The 10T SRAM cell has better read and write stabilitycompared to the 6T SRAM cell due to the additional access transistors.
- **Reduced leakage power:** The 10T SRAM cell reduces leakage power by using sleep transistors or power gating techniques.
- **Increased complexity:** The 10T SRAM cell is more complex than the 6T SRAM cell due to the additional transistors, which can increase the area and power consumption.
- **Compatibility:** The 10T SRAM cell can be used in both bulk CMOS and FinFET technologies.
- **Performance:** The 10T SRAM cell can achieve higher performance compared tothe 6T SRAM cell due to the reduced leakage power and improved stability.
- **Variability:** The 10T SRAM cell is more susceptible to process variation compared to the 6T SRAM cell due to the additional transistors.
- **Radiation hardness:** The 10T SRAM cell is designed to provide radiation hardness up to the value of LET = 35 MeV cm2/mg in 32-nm CMOS technology using TCAD simulations.

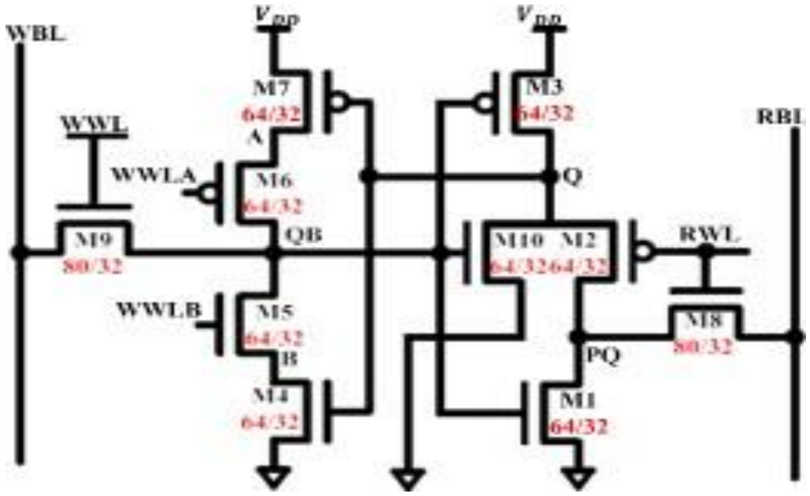


Fig 4.3 Schematic of 10T

4.3.1Power- Delay Products in Cmos:

The power delay product (PDP) is a power dissipation product and a delay of propagation. The following figure displays the waveform of PDP data.

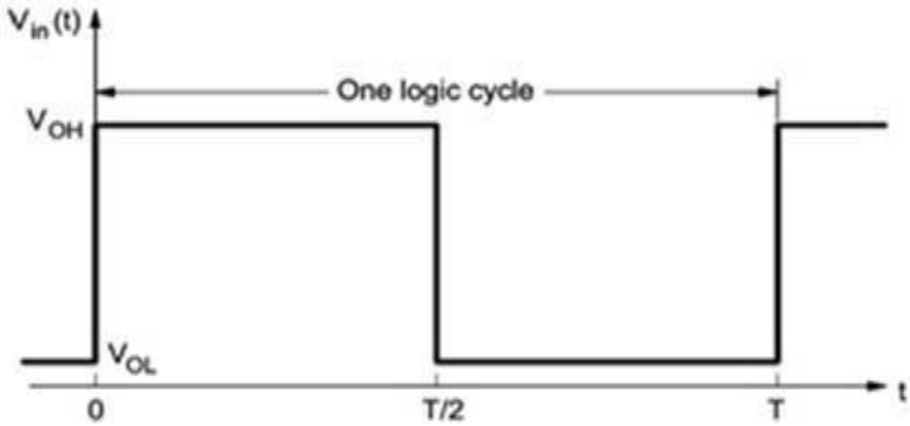


Fig. (A). Ideal Square Wave

Fig.(B). Finite Rise and Fall Time Waveform

Fig 4.4: Power Delay Product in CMOS

- PDP = Pav tp

The delay and power product (PDP) is an output voltage product and a delay ofpropagation. The following figure displays the waveform of PDP data.

- PDP = CLV2DD fmax tp = CLV2DD2

4.3.2 Power Dissipation Minimization Techniques:

Three methods are implemented to eliminate dissipation of power in digitally integratedcircuits

Scaling Voltage

Drop and clock size

Reduction of shifted power

- The input voltage is popular in stress scaling
- Since voltage drop and reduced power dissipation rely on power supply, power dissipation is decreased by decreased energy supply voltage
- The frequency of the clock processors is decreased by clock declining technology
- Higher clock frequency does not raising energy effectively. Nevertheless, when the clock signal is not used in ways that minimize resource dislocation because processors have power down ways

4.3.3 Total Power Consumption of Cmos Inverter:

Then the total energy dissipation requires into account all elements of output power,
I.e. the hydraulic pressure dissipation, evaporation in the specific current direction andthe power supply dissipation.

Ptotal = Pdynamic + Pdp + Pstatic

VDD = CL VDD2 f + tdp * VDD * Ip * f + Ileakage

VDD = (CLVDD2 + tdp * VDD * IP) *f + Ileakage

4.3.4 Static Power Consumption:

Thanks to power dissipation, the stagnant strength dissipation. The dielectric breakdown of the circuit's static or transient state strength,

- Pstat = Ileakage * VDD

Where I leakage is the movement of leakage from the VDD to the ground while switching is not carried out. Ideal for the pMOS and nMOS devices to be worked at the same time, the leak current of both the CMOS inverters is small. Nevertheless, the leakage current flows via transistor reversal diode contacts around sources or drain and ground. This new commitment is quite limited and should be disregarded. Consequently, CMOS inverters have very little and typically neglect the contribution of the static power dissipation portion.

4.3.5 Near- Threshold And Sub-Threshold Logic:

Engineers operating on a VLSI low power system considerably reduce the power supply voltages to a threshold voltage point or even below that of the circuit. The aim isto allow the most use of the equations relation between power usage and voltage: electricity is proportional to CV2f.

The power required for the transistor to adjust rapidly decreases as the voltage declines. This decrease is therefore followed by a drastic decrease in results. A further concern is the substantial rise in electricity that has been wasted owing to leakage as the voltage is lowered to below the acceptable maximum voltage.

In addition, the transistor leakage happens when the sub-threshold operator begins increasing the downstreamlogic's power. The increase in the leakage capacity contribution arises from the significant rise in the transistor switching period.

4.3.6 Sub-Threshold Vs Near-Threshold Logic:

Sub-threshold and near-threshold logic are techniques used in low-power VLSI design with reduced supply voltages. Sub-threshold logic operates below the threshold voltage, while near-threshold logic operates slightly above. Both methods aim to reduce power consumption, but they have different implications.

- **Sub-threshold logic**: This method significantly lowers power consumption but suffers from reduced performance and increased leakage due to the slow switching of transistors. It is more susceptible to process variation, making design more challenging.
- **Near-threshold logic:** This technique operates closer to the threshold voltage, balancing power consumption and performance. It is less susceptible to process variation compared to sub-threshold logic, making it more suitable for many low-power applications.
- **Process variation:** This is a critical factor in sub-threshold and near-threshold logic design. It refers to the differences between transistors, which can affect their behavior. Designers must account for this variation, making the design process more specialized and complex.
- **Voltage and current:** A significant difference between current levels for logic 1 and 0 is essential for proper signal discrimination. This difference ensures that thecircuit can accurately distinguish between the two logic states.